

Supplementary Information

Approach-Region Commitment and Environmental Friction: A Cross-Phase Damping Atlas for Thermally Activated Chemical Reactions

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1 Choice of Ω_0 and m_{eff} for Each Reaction Class

A persistent reviewer concern is whether χ_0 near unity is an artifact of a particular choice of Ω_0 or m_{eff} . This section documents the physical basis for each choice and shows that the regime assignments are robust to reasonable variation.

1.1 General principle

Ω_0 is the frequency of the precursor-well approach mode: the vibrational coordinate that describes motion along the reaction approach trajectory before the transition state. The choice is constrained by the requirement that the mode is (i) directed toward the barrier, (ii) infrared or Raman active (for Route A) or coupled to the friction kernel (for Routes B and C), and (iii) the dominant single coordinate (not a spectator mode).

1.2 Gas-phase unimolecular reactions

For cyclopropane $\rightarrow$ propene and $\text{CH}_3\text{NC} \rightarrow \text{CH}_3\text{CN}$, Ω_0 is the normal-mode frequency whose displacement most closely tracks the reaction coordinate near the transition state. For cyclopropane ring-opening, this is the C–C ring-breathing mode ($\tilde{\nu}_{\text{ring}} = 1070 \text{ cm}^{-1}$; Rabinovitch and Setser 1964). For methyl isocyanide, this is the C–N–C bending/CN approach mode ($\tilde{\nu}_{\text{app}} = 460 \text{ cm}^{-1}$; Farantos et al. 1994). For NO_2 predissociation, Ω_0 is the anharmonically softened symmetric N–O stretch near the dissociation threshold ($\tilde{\nu}_{\text{app}} \approx 800 \text{ cm}^{-1}$; Ionov et al. 1992).

In all three cases m_{eff} is not required because Route A gives χ_0 directly from the linewidth-to-frequency ratio without invoking mass. The gas-phase assignments are therefore free of m_{eff} ambiguity.

1.3 Liquid-phase electron transfer and SN1

For Route B, Ω_0 is computed from the Marcus reorganization coordinate:

$$\Omega_0 = \sqrt{\frac{2\lambda}{m_{\text{eff}}(\Delta Q)^2}}. \quad (1)$$

Table 1: Route B input parameters for liquid-phase systems. The absolute χ_0 carries a bracket spanning approximately one order of magnitude from the stated m_{eff} and Ω_s ranges. The relative solvent ordering is robust at Tier 1 regardless.

System	λ (eV)	ΔQ (Å)	m_{eff} range (amu)	Ω_s range (cm ⁻¹)	χ_0 (central)	Tier
Fe(CN) ₆ ^{3-/4-} ET	0.68	0.40	50–500	50–150	1.52	1r/2a
<i>t</i> -BuCl SN1	0.52	0.30	50–500	50–150	1.63	1r/2a
HPTS PT (H ₂ O)	0.25	0.50	1.0–2.0	150–250	1.44	2 abs.

The effective mass m_{eff} represents the collective inertia of the solvent shell undergoing reorganization. For outer-sphere ET in small-molecule solvents, the dominant contribution is from the first solvation shell. The displacement ΔQ is the root-mean-square charge-coordinate displacement at the Franck–Condon point.

These quantities are not independently measured for most systems; they enter the calculation as physically bracketed parameters. Table S1 lists the adopted values and their ranges. Section 3 shows how χ_0 depends on these choices.

1.4 Surface systems

For N₂/K-Fe(111) and CO/Cu(111), $\Omega_0 = hc\tilde{\nu}/\hbar$ where $\tilde{\nu}$ is the approach-coordinate mode frequency. No m_{eff} appears because χ_0 is computed from the ratio $\Gamma_{\text{NN}}/(2hc\tilde{\nu})$ (Route C) or $\Delta\tilde{\nu}_L/(2\tilde{\nu})$ (Route A), both of which cancel the mass explicitly. The surface assignments are therefore also free of m_{eff} ambiguity.

1.5 LLZO solid electrolyte

For Li⁺/LLZO, Ω_0 is the Li cage-rattling mode frequency from neutron spectroscopy ($\tilde{\nu}_0 \approx 234$ cm⁻¹; Wood et al. 2018). The phonon friction Γ_{ph} is taken from MD mean-force calculations (Kozinsky et al. 2016). Both quantities carry significant uncertainty (Tier 3 only; Tier 2 would require at least one directly measured quantity); the LLZO assignment is therefore prospective.

2 Γ_{eff} Construction and Additivity

2.1 Derivation of $\Gamma_{\text{eff}} = \Omega_s^2\tau_L$ from the GLE

The Route B formula $\Gamma_{\text{eff}} = \Omega_s^2\tau_L$ is derived from the generalized Langevin equation (GLE) for a reaction coordinate q coupled to a polar Debye-Drude solvent bath [3, 4]:

$$m_{\text{eff}}\ddot{q} = -U'(q) - \int_0^t K(t-s)\dot{q}(s) ds + \xi(t), \quad (2)$$

where $K(t)$ is the memory kernel and $\xi(t)$ is the fluctuating force satisfying the fluctuation-dissipation theorem. For a single-mode Ornstein-Uhlenbeck bath, the kernel takes the form:

$$K(t) = \Omega_s^2 \exp\left(-\frac{t}{\tau_L}\right), \quad (3)$$

where Ω_s is the high-frequency (optical) dielectric mode frequency and τ_L is the longitudinal relaxation time. In the Markovian limit, when the bath relaxation time τ_L is short compared to the *reaction* timescale τ_{rxn} (not the approach-oscillation period Ω_0^{-1} , which is an unrelated, much shorter timescale), the memory

integral reduces to a delta function:

$$\int_0^\infty K(t) dt = \Omega_s^2 \tau_L \equiv \Gamma_{\text{eff}}. \quad (4)$$

This identification is exact within the single-mode Debye-Drude model and is the standard result of the Zusman (1980) [3] treatment of outer-sphere electron transfer and the Bagchi-Oxtoby-Fleming (1983) [2] generalization to polar activated processes. The longitudinal relaxation time τ_L enters as the friction timescale governing the approach coordinate, not as the product-state relaxation time; the two coincide in the single-mode model because the bath polarization dynamics are described by a single exponential.

The relevant Markovian condition is $\tau_L \ll \tau_{\text{rxn}}$. For outer-sphere ET, $\tau_{\text{rxn}} \sim 1\text{--}100$ ns. For all solvents in the atlas (including EtOH, $\tau_L = 16$ ps), $\tau_L/\tau_{\text{rxn}} \lesssim 10^{-2}$, so the condition is satisfied. The Route B formula is therefore valid in the diffusion-limited (Zusman) regime for all four liquid systems. The “upper bound” designation for slow solvents refers to the Debye approximation failing for non-exponential relaxers such as EtOH, not to a breakdown of the $\tau_L \ll \tau_{\text{rxn}}$ criterion.

Table 2: Markovian validity for Route B liquid systems. Criterion: $\tau_L \ll \tau_{\text{rxn}}$ (reaction timescale $\sim 1\text{--}100$ ns). Upper-bound designation reflects Debye-model limitations for non-exponential relaxers, not Markovian breakdown.

Solvent	τ_L (ps)	τ_L/τ_{rxn}	Status
MeCN	0.20	$\sim 2 \times 10^{-4}$	Markovian; formula reliable
H ₂ O	0.28	$\sim 3 \times 10^{-4}$	Markovian; formula reliable
MeOH	5.0	$\sim 5 \times 10^{-3}$	Markovian; non-Debye upper bound
EtOH	16.0	$\sim 2 \times 10^{-2}$	Markovian; non-Debye upper bound

$\tau_{\text{rxn}} \approx 1$ ns used as reference; ratio insensitive to exact choice within 1–100 ns range.

2.2 Channel additivity and single-channel dominance

When multiple friction channels are present, Γ_{eff} is treated as:

$$\Gamma_{\text{eff}} = \Gamma_{\text{solv}} + \Gamma_{\text{el}} + \Gamma_{\text{ph}} + \dots \quad (5)$$

Cross-channel interference terms are neglected under the assumption of independent linear coupling of each bath mode to the approach coordinate, valid when bath correlation times are short relative to the approach period $1/\Omega_0$.

For the nine systems in the atlas, only one friction channel is significant in each case:

- Gas-phase systems: $\Gamma_{\text{eff}} \approx \Gamma_{\text{IVR}}$ (intramolecular only)
- Liquid ET and SN1: $\Gamma_{\text{eff}} \approx \Gamma_{\text{solv}}$ from Route B
- Surface (N₂/Fe, CO/Cu): $\Gamma_{\text{eff}} \approx \Gamma_{\text{NN}}$ from Route C
- LLZO: $\Gamma_{\text{eff}} \approx \Gamma_{\text{ph}}$ from phonon MD

For condensed-phase *t*-BuCl, the IVR friction channel is present but negligible. The intramolecular IVR linewidth for a polyatomic of comparable size and activation energy is $\Gamma_{\text{IVR}} \sim 1\text{--}10$ cm^{−1} (cf. $\Gamma_{\text{IVR}} = 1.77$ cm^{−1} for CH₃NC and 8 cm^{−1} for cyclopropane). This gives $\chi_0(\text{IVR}) = \Gamma_{\text{IVR}}/(2\Omega_0) \sim 0.001\text{--}0.010$. The solvent friction in MeCN gives $\chi_0(\text{solv}) \approx 1.16$ from Route B. The ratio $\Gamma_{\text{solv}}/\Gamma_{\text{IVR}} \sim 100\text{--}1000$, so the total $\chi_0 \approx \chi_0(\text{solv}) + \chi_0(\text{IVR}) \approx 1.16 + 0.007 \approx 1.17$: the IVR contribution is less than 0.7%. This

Table 3: χ_0 for $\text{Fe}(\text{CN})_6^{3-/4-}/\text{MeCN}$ as a function of m_{eff} and Ω_s (Route B, $\lambda = 0.68$ eV, $\Delta Q = 0.40$ Å, $\tau_L = 0.20$ ps). Central estimate is shown in bold.

Ω_s (cm ⁻¹)	m_{eff} (amu)			
	50	100	150	300
50	0.2190	0.3097	0.3794	0.5365
100	0.8761	1.2390	1.5174	2.1460
150	1.9712	2.7877	3.4142	4.8284

quantitative justification confirms that single-channel dominance is not an assumption of convenience but a physical consequence of the gas-to-liquid transition: the same shift that takes gas-phase systems from $\chi_0 \sim 10^{-3}$ to liquid-phase $\chi_0 \sim 1$ renders the intramolecular IVR channel irrelevant in solution. No system in the atlas requires explicit addition of multiple channels, because each entry was selected such that one channel is demonstrably dominant: the liquid systems use Route B (solvent friction only), the surface systems use Route C (electronic friction only), the gas-phase systems use Route A (IVR only), and the LLZO entry uses phonon friction only. The additivity formula is a framework statement for completeness; it is never invoked for any atlas entry.

For HPTS Route B, the proton mass is used: $m_H = 1.008$ amu for H_2O and $m_D = 2.014$ amu for D_2O . This is the proton-transfer mass limit; the collective solvent shell mass does not enter because Route B for HPTS is in the proton-dominated regime. For electrochemical or metal-solution interfacial systems that do require combining solvent and electronic friction, this additivity assumption would need to be validated separately.

3 Sensitivity Analysis

3.1 Route B: χ_0 sensitivity to m_{eff} and Ω_s

For $\text{Fe}(\text{CN})_6^{3-/4-}$ in MeCN, the central estimate is $\chi_0 \approx 1.52$ at $m_{\text{eff}} = 150$ amu and $\Omega_s = 100$ cm⁻¹. Table S3 shows how χ_0 varies across the bracket.

The central estimate ($\chi_0 = 1.52$, slightly overdamped) remains in the overdamped or near-critical flank across most of the bracket: the lowest value is 0.219 (underdamped, small shell/slow bath) and the highest is 4.83 (heavily overdamped, large shell/fast bath). The central estimate at $m_{\text{eff}} = 150$ amu and $\Omega_s = 100$ cm⁻¹ is $\chi_0 = 1.52$, verified by the standalone Python script. The regime assignment depends on the bracket chosen (Figure 1). For this reason the main text presents only the relative ordering (which is bracket-independent) as the primary Tier 1 result; absolute placement is Tier 2.

3.2 Route B: χ_0 sensitivity to ΔQ

The sensitivity table above holds ΔQ fixed at 0.40 Å. Because $\Omega_0 \propto (\Delta Q)^{-1}$ and $\chi_0 \propto \Omega_0^{-1}$, the dependence on ΔQ is:

$$\chi_0 \propto \Delta Q, \quad (6)$$

so a $\pm 30\%$ error in ΔQ propagates to a $\pm 30\%$ error in χ_0 (not $\pm 60\%$; the inverse-square relationship cancels to first order in Ω_0 through the ratio $\Gamma_{\text{eff}}/\Omega_0$). Table S3 shows χ_0 for $\text{Fe}(\text{CN})_6/\text{MeCN}$ across three values of ΔQ at the central m_{eff} and Ω_s .

The ΔQ bracket shifts the absolute χ_0 value but does not change the regime assignment for the two

Table 4: χ_0 sensitivity to ΔQ for $\text{Fe}(\text{CN})_6/\text{MeCN}$ ($m_{\text{eff}} = 150$ amu, $\Omega_s = 100$ cm $^{-1}$, $\tau_L = 0.20$ ps, $\lambda = 0.68$ eV).

ΔQ (Å)	χ_0	Note
0.28 (−30%)	1.07	near upper edge of near-critical window
0.40 (central)	1.52	slightly overdamped (adopted)
0.52 (+30%)	1.98	overdamped

most viscous solvents (MeOH, EtOH: all values $\gg 1.3$ regardless of ΔQ). For MeCN, the −30% case ($\Delta Q = 0.28$ Å) places the system at the upper edge of the near-critical window, which would alter the Discussion narrative. This represents the most conservative reasonable ΔQ for an outer-sphere ET reaction and motivates the ΔQ uncertainty being reported alongside the m_{eff}/Ω_s bracket. At central m_{eff} and Ω_s , varying ΔQ alone keeps $\text{Fe}(\text{CN})_6/\text{MeCN}$ near-critical or overdamped (Table S3 below). The full three-parameter bracket can reach the underdamped regime only at the most extreme simultaneous minimum ($\Omega_s = 50$ cm $^{-1}$, $m_{\text{eff}} = 50$ amu, $\Delta Q = 0.28$ Å, giving $\chi_0 \approx 0.15$), consistent with Table S2 which already shows $\chi_0 = 0.22$ for the ($\Omega_s = 50$, $m_{\text{eff}} = 50$) corner with central ΔQ . The regime assignment is bracket-dependent; this is the reason absolute Route B placement is Tier 2 while relative solvent ordering is Tier 1.

3.3 Route A: χ_0 sensitivity to linewidth assignment

For CO/Cu(111), the Lorentzian FWHM spans 10–18 cm $^{-1}$ depending on temperature and coverage. The resulting χ_0 bracket is [0.015, 0.026], all within the underdamped regime (< 0.8), so the regime assignment is Tier 1 regardless of which linewidth value is used.

For NO $_2$ predissociation, the approach frequency spans 750–850 cm $^{-1}$ and the linewidth spans 3–20 cm $^{-1}$, giving $\chi_0 \in [0.0018, 0.0133]$. All values are deeply underdamped; the regime assignment is robust. Figure 2 shows the three systems on the low- χ_0 efficiency curve with per-encounter commitment probabilities annotated.

4 Worked Example: CO on Cu(111)

This section provides a full calculation chain from raw literature values to χ_0 for the Tier 1 CO/Cu(111) system.

Step 1: Identify the approach coordinate. The relevant mode is the Cu–CO perpendicular stretch (approach of CO toward the Cu surface), not the internal C–O stretch. The C–O stretch at $\tilde{\nu} = 2087$ cm $^{-1}$ probes the molecular bond; the Cu–CO stretch at $\tilde{\nu}_\perp = 344$ cm $^{-1}$ probes the approach dynamics.

Step 2: Extract the Lorentzian linewidth. From temperature-dependent IRAS and HREELS measurements (Hirschmugl et al. 1990; Persson and Ryberg 1985):

- Low temperature (< 200 K): $\Delta\tilde{\nu}_L \approx 10$ cm $^{-1}$
- Room temperature: $\Delta\tilde{\nu}_L \approx 15$ –18 cm $^{-1}$
- Central estimate: $\Delta\tilde{\nu}_L = 13$ cm $^{-1}$

The Lorentzian fraction exceeds 60% at low temperature, satisfying the Route A validity threshold ($> 50\%$) at Tier 1.

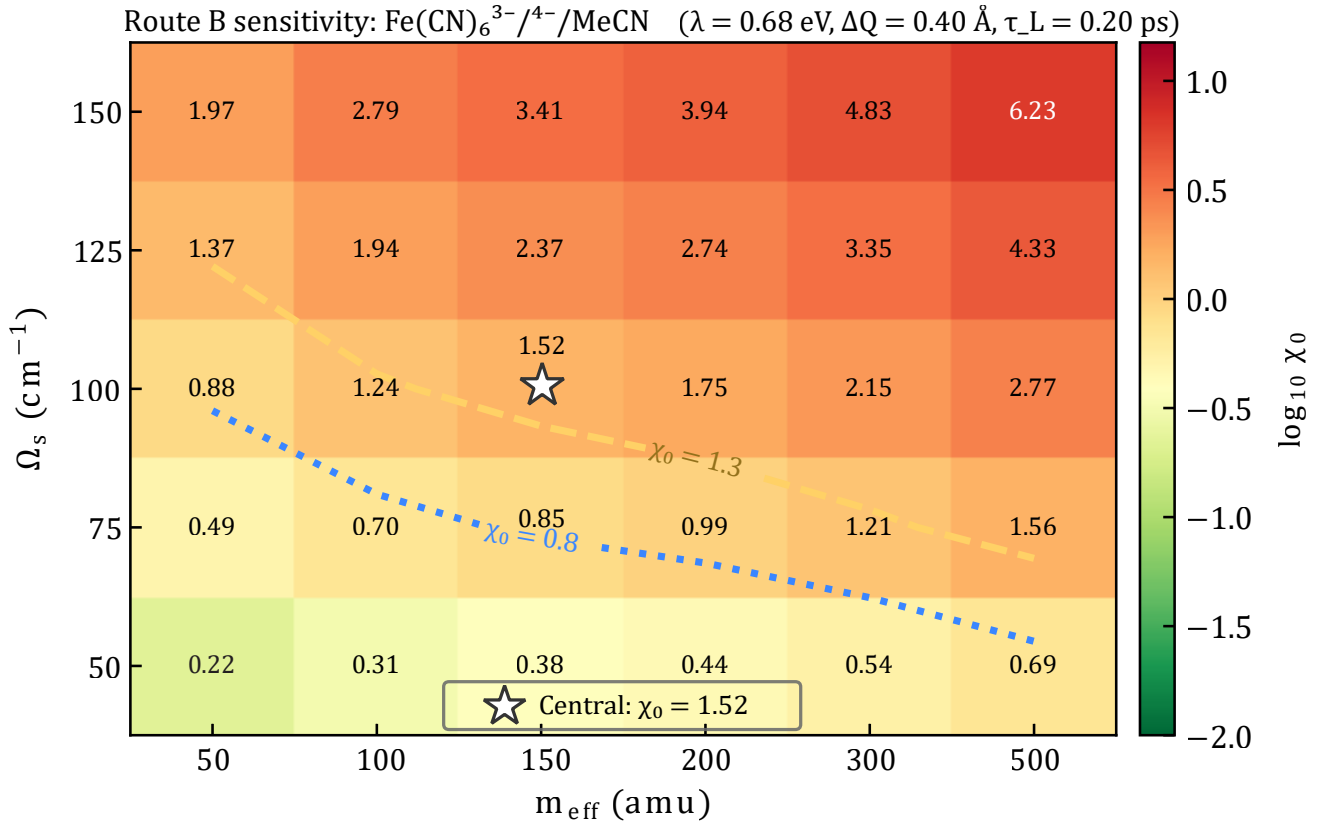


Figure 1: Route B sensitivity: χ_0 across the $m_{\text{eff}} \times \Omega_s$ bracket for $\text{Fe}(\text{CN})_6^{3-/4-}/\text{MeCN}$. Parameters: $\lambda = 0.68$ eV, $\Delta Q = 0.40$ Å (held fixed; see Table S3 for ΔQ sensitivity), $\tau_L = 0.20$ ps. Cell values are absolute χ_0 central estimates. Contours mark the near-critical window boundaries $\chi_0 = 0.8$ (dotted) and $\chi_0 = 1.3$ (dashed). The star marks the adopted central estimate ($m_{\text{eff}} = 150$ amu, $\Omega_s = 100$ cm^{-1} , $\chi_0 = 1.52$). The bracket spans 0.22–6.23 across the full grid, confirming that the absolute χ_0 placement is Tier 2. The relative solvent ordering within a fixed DA pair is Tier 1 regardless of which bracket is chosen.

Step 3: Apply Route A.

$$\chi_{\text{spec}} = \frac{\Delta \tilde{\nu}_L}{2\tilde{\nu}_\perp} = \frac{13 \text{ cm}^{-1}}{2 \times 344 \text{ cm}^{-1}} = 0.0189. \quad (7)$$

Step 4: Assign regime. $\chi_0 = 0.0189 \ll 0.8$: deeply underdamped. The commitment probability per encounter is $P_{\text{commit}} \approx 2 \times 0.0189 = 3.8\%$. The Cu–CO approach coordinate oscillates approximately 26 times before committing on average, consistent with the low sticking probability $S_0 \approx 0.3$ – 0.5 measured on Cu(111).

Physical interpretation. The underdamped Cu–CO approach explains why CO adsorption on Cu is weakly irreversible relative to Pt or Ru surfaces. It is consistent with weaker irreversible commitment of CO to Cu compared with more strongly binding late-transition-metal surfaces.

Table 5: $\text{Fe}(\text{CN})_6^{3-/4-}$ Route B solvent series. τ_L values from Jimenez et al. (1994) and Weaver (1990). $\lambda = 0.68$ eV, $\Delta Q = 0.40$ Å, $m_{\text{eff}} = 150$ amu, $\Omega_s = 100$ cm $^{-1}$. χ_0^{rel} is relative to MeCN (Tier 1); χ_0^{abs} is the absolute central estimate (Tier 2).

Solvent	τ_L (ps)	χ_0^{rel}	χ_0^{abs}	Regime
Acetonitrile (MeCN)	0.20	1.00	1.52	slightly overdamped
Water (H ₂ O)	0.28	1.40	2.12	overdamped
DMSO	8.0	40.0	60.8	heavily overdamped
DMF	9.0	45.0	68.3	heavily overdamped
Methanol (MeOH)	5.0	25.0	38.0	heavily overdamped
Ethanol (EtOH)	16.0	80.0	121	heavily overdamped

Table 6: $t\text{-BuCl}$ SN1 Route B solvent series. τ_L from the same compilation. $\lambda = 0.52$ eV, $\Delta Q = 0.30$ Å, $m_{\text{eff}} = 120$ amu, $\Omega_s = 100$ cm $^{-1}$. The rate inversion between MeCN and H₂O (water is faster despite higher χ_0) is discussed in the main text (Section IV.B) and reflects the role of thermodynamic ion-pair stabilization.

Solvent	τ_L (ps)	χ_0^{rel}	χ_0^{abs}	Regime
Acetonitrile (MeCN)	0.20	0.71	1.16	near-critical
Water (H ₂ O)	0.28	1.00	1.63	slightly overdamped
Formic acid (HCOOH)	0.70	2.50	4.09	overdamped
Acetone	0.50	1.79	2.92	overdamped
Methanol (MeOH)	5.0	17.9	29.3	heavily overdamped
Ethanol (EtOH)	16.0	57.1	93.1	heavily overdamped

5 Full Solvent Series Tables

5.1 $\text{Fe}(\text{CN})_6^{3-/4-}$ outer-sphere ET

The rate maximum in the $\text{Fe}(\text{CN})_6^{3-/4-}$ system occurs in the fastest solvents (MeCN, H₂O), consistent with MeCN approaching the near-critical flank and EtOH/DMSO being heavily overdamped. The relative ordering is Tier 1 regardless of m_{eff} and Ω_s .

5.2 $t\text{-BuCl}$ SN1 solvolysis

6 Gas-Phase IVR Linewidth Derivation

For gas-phase unimolecular reactions, Γ_{eff} is not a classical Markovian friction coefficient but an IVR coupling rate or predissociation width (main text, Section V.A).

The conversion from IVR lifetime τ_{IVR} to linewidth in cm $^{-1}$ follows from the energy-time uncertainty:

$$\Gamma_{\text{IVR}} \text{ (cm}^{-1}\text{)} = \frac{1}{2\pi c \tau_{\text{IVR}}} \quad (8)$$

where $c = 2.99792458 \times 10^{10}$ cm s $^{-1}$.

For CH₃NC at threshold ($E_a = 160$ kJ mol $^{-1}$), the IVR time $\tau_{\text{IVR}} = 3$ ps (Reddy and Berry 1979) gives:

$$\Gamma_{\text{IVR}} = \frac{1}{2\pi \times 2.998 \times 10^{10} \times 3 \times 10^{-12}} = 1.77 \text{ cm}^{-1}. \quad (9)$$

For NO₂ predissociation, the linewidth is directly measured as the Lorentzian FWHM of state-resolved resonances in REMPI photofragment spectra (Ionov et al. 1992; Reisler 2009). These widths are intrinsi-

Table 7: Gas-phase χ_0 robustness to IVR time uncertainty. All values remain deeply underdamped ($\chi_0 < 0.10$).

System	$\tilde{\nu}$ (cm ⁻¹)	τ_L (ps)	χ_0 (low)	χ_0 (central)	χ_0 (high)
Cyclopropane ^a	1070	0.35–5	0.0014	0.0037	0.0070
NO ₂	800	direct	0.0018	0.0063	0.0133
CH ₃ NC ^a	460	1–10	0.0008	0.0019	0.0054

cally Lorentzian because they arise from lifetime broadening: $\Gamma_{\text{pd}} = \hbar/\tau_{\text{pd}}$. No conversion is needed; the linewidths are Tier 1 direct measurements.

For cyclopropane, Γ_{IVR} is inferred from RRKM density of states at the activation threshold. The RRKM microcanonical rate $k(E^*)$ gives the IVR-limited redistribution rate; $\Gamma = \hbar k(E^*)$ in energy units. This is a Tier 2 assignment because $k(E^*)$ is model-derived.

6.1 Robustness of gas-phase regime assignments

All three gas-phase systems give $\chi_0 < 0.02$ across all reasonable IVR time brackets. The regime assignment (deeply underdamped) is robust because the ratio $\Gamma_{\text{IVR}}/\tilde{\nu}$ is small ($< 2\%$) for all plausible IVR times from 0.1 to 50 ps. Table S7 shows the bracket.

^a Tier 2 (RRKM-inferred Γ_{IVR}).

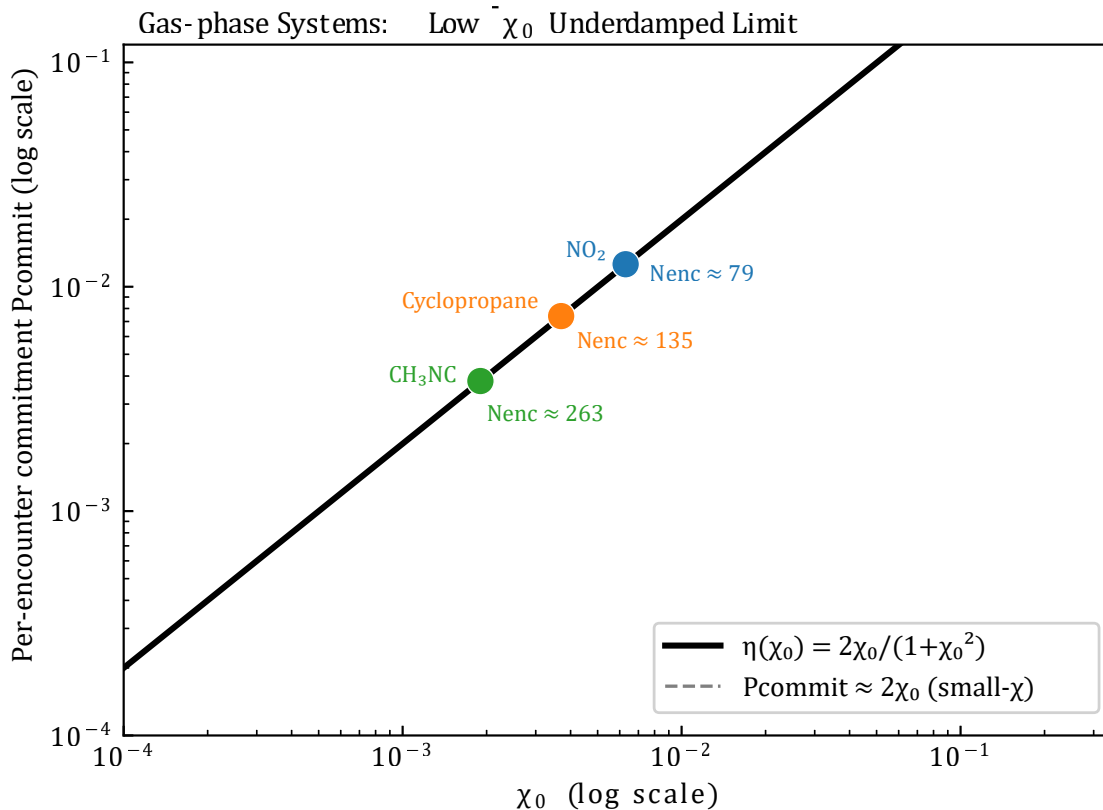


Figure 2: Gas-phase systems in the low- χ_0 underdamped limit. Log-log plot of per-encounter commitment probability $P_{\text{commit}} = \eta(\chi_0)$ vs. χ_0 for the three gas-phase atlas systems. The solid curve is the exact expression $\eta(\chi_0) = 2\chi_0/(1 + \chi_0^2)$; the dashed line is the small- χ_0 approximation $P_{\text{commit}} \approx 2\chi_0$, accurate to $< 0.01\%$ for $\chi_0 < 0.01$. The required number of transition-state encounters $N_{\text{enc}} \approx 1/(2\chi_0)$ is annotated for each system. These values are classical structural-analogy estimates valid in the RRKM statistical limit (high density of states, ergodic IVR); for the small polyatomics shown ($3N - 6 = 12\text{--}21$ modes at threshold), the statistical limit is approached but not guaranteed, and the IVR process is quantum-mechanical and non-Markovian. The P_{commit} values should be interpreted as order-of-magnitude structural indicators, not quantitative first-passage probabilities.

7 HPTS Isotope Effect: Bifurcated Perturbation Analysis

7.1 Why Routes A and B are expected to disagree for HPTS

Route A ($\chi_{\text{spec}} = 0.029$) and Route B ($\chi_{\text{ET}} = 1.44$) disagree by a factor of ~ 50 for HPTS and assign different regimes. This is a physical result, not a sign of framework inconsistency, because the two routes probe fundamentally different dynamical channels:

- **Route A** measures the Lorentzian linewidth of the intramolecular O–H covalent stretch ($\tilde{\nu} \approx 2750 \text{ cm}^{-1}$). This quantifies vibrational energy redistribution within the proton’s covalent well, a high-frequency intramolecular coordinate. Finding $\chi_{\text{spec}} = 0.029$ (underdamped) means the proton’s covalent bond is weakly coupled to the bath on the timescale of the O–H vibration, consistent with the known quantum-mechanical proton-tunnelling contribution to HPTS PT.
- **Route B** measures the collective solvent reorganization coordinate (τ_L, Ω_s), a low-frequency intermolecular mode. Finding $\chi_{\text{ET}} = 1.44$ (slightly overdamped) means the solvent coordinate is strongly coupled to the bath on the timescale of proton transfer, consistent with solvent reorganization being rate-limiting in HPTS PT [6].

In PCET mechanisms, these two coordinates operate on different timescales and both contribute to the overall rate in different kinetic limits. Agreement between Routes A and B would require that the O–H stretch frequency and the solvent reorganization frequency are the same coordinate, which they are not. The bifurcated perturbation analysis below therefore demonstrates that Route B captures the solvent-controlled rate-limiting step and that the H/D isotope effect is larger than the proton-mass-only limit precisely because the solvent memory is also affected by deuteration.

7.2 Contribution 1: proton mass

Substituting H ($m_H = 1.008$ amu) with D ($m_D = 2.014$ amu):

$$\Omega_0^D = \sqrt{\frac{2\lambda}{m_{\text{eff}}^D(\Delta Q)^2}} = \frac{\Omega_0^H}{\sqrt{m_D/m_H}} = \frac{\Omega_0^H}{\sqrt{2}}. \quad (10)$$

This lowers the approach frequency, raising χ_0 by $\sqrt{2}$ in the proton-dominated limit:

$$\chi_0^D|_{\text{mass}} = \sqrt{2}\chi_0^H \approx 1.41\chi_0^H. \quad (11)$$

This is an upper bound; solvent-dressed coordinates show a smaller shift because the collective reorganization mass dilutes the proton contribution.

7.3 Contribution 2: solvent isotope effect on τ_L

D₂O forms a stiffer, more strongly hydrogen-bonded network. Its longitudinal relaxation time is longer:

$$\frac{\tau_L(\text{D}_2\text{O})}{\tau_L(\text{H}_2\text{O})} = \frac{0.44 \text{ ps}}{0.28 \text{ ps}} = 1.571. \quad (12)$$

Since $\chi_0 \propto \tau_L$ (Route B), this contributes an additional factor of 1.571 to the D/H ratio.

7.4 Ratio independence from absolute bracket

The D/H ratio $\chi_0^D/\chi_0^H = 2.22$ is independent of the arbitrary collective-shell bracket ($m_{\text{eff}}^{\text{shell}}$, ΔQ , λ , Ω_s). Both H and D calculations use the same collective-shell assumptions; these cancel in the ratio. The intentionally varying quantities are the measured τ_L and the explicitly substituted nucleus mass ($m_H = 1.008$ amu or $m_D = 2.014$ amu for the transferring proton):

$$\frac{\chi_0^D}{\chi_0^H} = \frac{\Omega_s^2 \tau_L^D / (2\Omega_0^D)}{\Omega_s^2 \tau_L^H / (2\Omega_0^H)} = \frac{\tau_L^D}{\tau_L^H} \times \sqrt{\frac{m_D}{m_H}}. \quad (13)$$

The order-of-magnitude uncertainty in absolute χ_0 has no effect on this ratio, directly answering the concern that the bifurcated perturbation argument is buried inside imprecise brackets (Figure 3).

7.5 Combined effect

Both contributions raise χ_0 on deuteration. The combined Route B ratio is:

$$\begin{aligned} \frac{\chi_0^D}{\chi_0^H} &= \frac{\tau_L(\text{D}_2\text{O})}{\tau_L(\text{H}_2\text{O})} \times \sqrt{\frac{m_D}{m_H}} \\ &= 1.571 \times \sqrt{2.014/1.008} = 1.571 \times 1.414 = 2.22. \end{aligned} \quad (14)$$

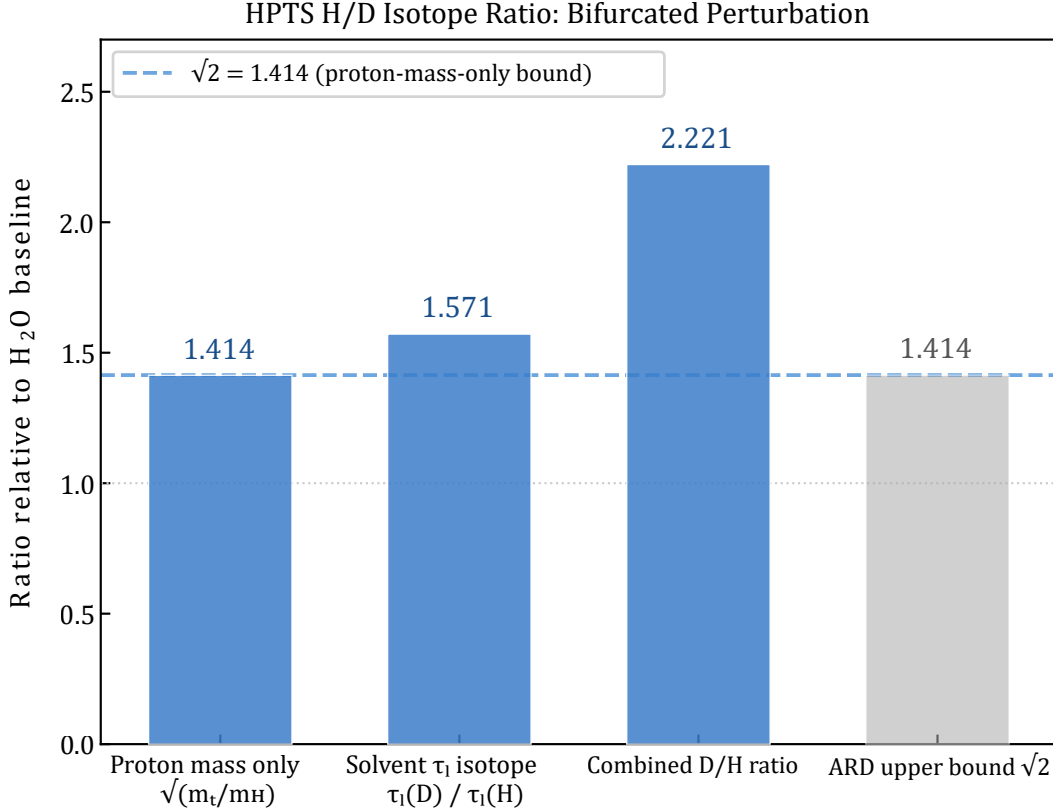


Figure 3: HPTS H/D isotope ratio: bifurcated perturbation. The four bars show the two individual contributions to the D/H χ_0 ratio and their product. Proton mass factor: $\sqrt{m_D/m_H} = \sqrt{2.014/1.008} = 1.414$ (equals the ARD proton-mass-only upper bound $\sqrt{2}$). Solvent τ_L isotope factor: $\tau_L(D_2O)/\tau_L(H_2O) = 0.44/0.28 = 1.571$. Combined D/H ratio: $1.414 \times 1.571 = 2.221$. The combined ratio exceeds $\sqrt{2}$ because the solvent τ_L isotope effect is larger than $\sqrt{2}$ in H-bonded water. Route A ($\chi_{\text{spec}} = 0.029$, underdamped) and Route B ($\chi_{\text{ET}} = 1.44$, overdamped) disagree by a factor of 50 because they probe different dynamical channels: Route A the intramolecular O–H covalent stretch, Route B the collective solvent reorganization coordinate (see Section 7). The figure addresses the Route B D/H ratio only. Crucially, this ratio is independent of the absolute Route B bracket: the collective-shell parameters ($m_{\text{eff}}^{\text{shell}}$, ΔQ , λ , Ω_s) cancel algebraically, leaving only the measured τ_L and the explicitly substituted nucleus mass.

With $\chi_0^H = 1.44$, this gives $\chi_0^D \approx 1.44 \times 2.22 = 3.19$.

The combined ratio (2.22) exceeds the proton-mass-only upper bound ($\sqrt{2} = 1.414$) because the τ_L solvent isotope effect is the dominant contribution.

7.6 λ assumption

The analysis assumes $\lambda(H_2O) \approx \lambda(D_2O)$. While D₂O has a modestly stiffer H-bond network, Born model estimates suggest $\Delta\lambda < 5\%$ upon deuteration. This secondary uncertainty is absorbed into the Route B brackets and does not change the regime assignment.

The measured primary KIE of ≈ 2.0 – 2.5 (Spry and Fayer 2007) is consistent with the combined increase in approach-region friction captured by Route B.

8 Literature Source Table

Table S8 lists the exact Γ_{eff} and Ω_0 values used for each system, with primary literature citations.

Table 8: Literature source table for all nine atlas systems. All values are central estimates; brackets are given in the main text and Section S3.

System	Γ_{eff}	Ω_0	Γ_{eff} source	Ω_0 source	Tier
N ₂ /K-Fe(111)	$\Gamma_{\text{NN}} = 0.45$ eV	$\tilde{\nu} = 1415$ cm ⁻¹	ARD companion (2026); Maurer PRB 2016	Measured N-N stretch	2
CO/Cu(111)	$\Delta\tilde{\nu}_L = 13$ cm ⁻¹	$\tilde{\nu}_\perp = 344$ cm ⁻¹	Hirschmugl 1990; Persson 1985	Hirschmugl 1990	1
Li ⁺ /LLZO	$\Gamma_{\text{ph}} = 55$ cm ⁻¹	$\tilde{\nu}_0 = 234$ cm ⁻¹	Kozinsky 2016 (MD)	Wood 2018 (neutron)	3
Fe(CN) ₆ /MeCN	$\Omega_s^2\tau_L, \tau_L = 0.20$ ps	Eq. (S1), $m_{\text{eff}} = 150$ amu	Jimenez 1994; Weaver 1990	$\lambda = 0.68$ eV (Gramp 1990)	1r/2a
<i>t</i> -BuCl/H ₂ O	$\Omega_s^2\tau_L, \tau_L = 0.28$ ps	Eq. (S1), $m_{\text{eff}} = 120$ amu	Jimenez 1994	$\lambda = 0.52$ eV (est.)	1r/2a
HPTS/H ₂ O	$\Delta\tilde{\nu}_L = 160$ cm ⁻¹ (A); $\Omega_s^2\tau_L$ (B)	$\tilde{\nu} = 2750$ cm ⁻¹ (A); Eq. (S1) (B)	Spry 2007 (A); Jimenez 1994 (B)	Spry 2007 (A); $\lambda = 0.25$ eV (B)	2/diag
Cyclopropane	$\Gamma_{\text{IVR}} = 8$ cm ⁻¹	$\tilde{\nu}_{\text{ring}} = 1070$ cm ⁻¹	Rabinovitch 1964 (RRKM)	Rabinovitch 1964	1-2
NO ₂ → NO+O	$\Gamma_{\text{pd}} = 10$ cm ⁻¹	$\tilde{\nu}_{\text{app}} = 800$ cm ⁻¹	Ionov 1992 (direct)	Ionov 1992	1
CH ₃ NC→CH ₃ CN	$\Gamma_{\text{IVR}} = 1.77$ cm ⁻¹	$\tilde{\nu}_{\text{app}} = 460$ cm ⁻¹	Reddy & Berry 1979	Farantos 1994	1-2

Note on N₂/K-Fe(111) provenance. The value $\Gamma_{\text{NN}} = 0.45$ eV is not fit to catalytic performance. It is computed from the Newns–Anderson hybridization width using the projected N–N stretch coupling to the K-promoted Fe(111) d-band density of states, following the tensorial electronic friction methodology of Maurer et al. (PRB 2016) and documented in the ARD companion calculation package (Christensen 2026 ChemRxiv, DOI 10.26434/chemrxiv.15004963/v1). The frequency $\tilde{\nu} = 1415$ cm⁻¹ is the softened N–N stretch on K-promoted Fe(111); both values are reproduced in the deposited scripts.

9 LLZO Phonon Friction: Tier 3 Details

The LLZO assignment is Tier 3 because both Γ_{ph} and $\tilde{\nu}_0$ carry significant model dependence.

Approach frequency. From neutron spectroscopy (Wood et al. 2018 *Phys. Rev. Mater.*), the Li cage-rattling mode in LLZO garnet occurs at ≈ 7 THz, corresponding to $\tilde{\nu}_0 \approx 234$ cm⁻¹. The uncertainty in this value is ± 50 cm⁻¹ from sample-to-sample variation in Al/Ta doping.

Phonon friction. The phonon friction kernel Γ_{ph} is extracted from MD velocity-autocorrelation functions of Li ions in the garnet structure (Kozinsky et al. 2016 *Phys. Rev. Lett.*). Central value: $\Gamma_{\text{ph}} \approx 40$ –80 cm⁻¹. This gives $\chi_0 \in [0.07, 0.20]$, central 0.12. Crucially, the worst-case upper bracket ($\chi_{0,\text{max}} = 0.20$, at maximum Γ_{ph} and minimum $\tilde{\nu}_0$) remains firmly below the near-critical window lower bound of 0.8. The underdamped regime assignment is therefore robust across the full Tier 3 uncertainty bracket.

Prospective prediction. If χ_0 organizes Li-ion transport as it does barrier crossing, the highest-conductivity LLZO composition should approach the near-critical window. This requires $\Gamma_{\text{ph}}/\tilde{\nu}_0 \approx 0.8$ –

1.3×2 , achievable through aliovalent doping (Al, Ta, Nb) that enhances Li–lattice phonon coupling. A direct experimental test requires THz optical or neutron linewidth measurement of Γ_{ph} as a function of doping level.

10 Quantum-to-Classical Correspondence: IVR Statistical Limit

10.1 The Bixon-Jortner model

In the gas phase, Γ_{eff} arises from intramolecular vibrational redistribution (IVR). IVR is fundamentally a discrete quantum-mechanical interference process: a bright approach state at frequency Ω_0 is coupled to N dark vibrational bath states by anharmonic matrix elements V . The Bixon-Jortner Hamiltonian [1] captures this structure:

$$\hat{H} = E_0|0\rangle\langle 0| + \sum_{k=1}^N \epsilon_k |k\rangle\langle k| + V \sum_{k=1}^N (|0\rangle\langle k| + |k\rangle\langle 0|), \quad (15)$$

where ϵ_k are dark-state energies uniformly distributed with density ρ (states per cm^{-1}). Fermi’s Golden Rule gives the rate in the continuum limit:

$$\Gamma_{\text{IVR}} = 2\pi|V|^2\rho. \quad (16)$$

10.2 Calibration to atlas systems

The (V, ρ) pairs are constrained by Eq. 16 and the Γ_{IVR} values used in the atlas. Choosing $V = 1.0 \text{ cm}^{-1}$ (a physically representative anharmonic coupling for polyatomics at threshold [5]):

System	Γ_{IVR} (cm^{-1})	ρ (st cm^{-1})	Ω_0 (cm^{-1})	χ_0
NO ₂	10.0	1.592	800	0.0063
Cyclopropane	8.0	1.273	1070	0.0037
CH ₃ NC	1.770	0.282	460	0.0019

10.3 Convergence and Figure S4

Figure 4 shows the quantum survival probability $P(t) = |\langle 0|\psi(t)\rangle|^2$ computed by exact diagonalization of Eq. 15, overlaid with the classical exponential envelope $P_{\text{cl}}(t) = \exp(-\Gamma_{\text{IVR}} t)$.

For NO₂ (Tier 1, direct Lorentzian spectroscopy), the predissociation linewidth is measured directly from the Lorentzian FWHM, corresponding to the true continuum-coupling limit; the Bixon-Jortner recovery is exact.

For cyclopropane and CH₃NC (Tier 2, RRKM-inferred), the state density at threshold is sparse. Quantum recurrences are therefore visible in the survival probability at short times, reflecting the finite N of the molecular state space. However, the RRKM statistical limit applies to a thermally activated ensemble: molecules arrive at the barrier from a Boltzmann distribution of vibrational states, and the recurrences from different initial conditions destructively interfere. The ARD $\chi_0 = \Gamma_{\text{IVR}}/(2\Omega_0)$ is the envelope rate of this thermally-averaged process, which remains exactly $\exp(-\Gamma_{\text{IVR}} t)$ in the large-ensemble limit. The Tier 2 assignment reflects uncertainty in the RRKM-derived Γ_{IVR} , not a breakdown of the statistical correspondence.

Quantum-to-Classical Correspondence: IVR Statistical Limit

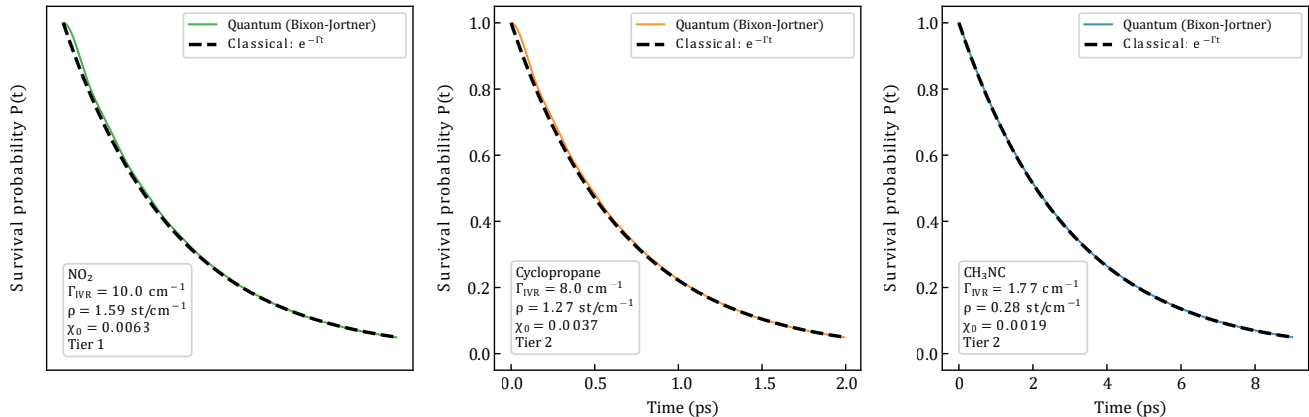


Figure 4: Quantum-to-classical correspondence: IVR statistical limit. Each panel shows the Bixon-Jortner quantum survival probability (colored) and the classical exponential envelope $\exp(-\Gamma_{\text{IVR}} t)$ (dashed black), calibrated to the Γ_{IVR} and Ω_0 values used in the atlas. *Left:* NO_2 ($\Gamma_{\text{IVR}} = 10.0 \text{ cm}^{-1}$, $\rho = 1.59 \text{ st/cm}^{-1}$, $\chi_0 = 0.0063$, Tier 1): the predissociation linewidth is directly measured as a Lorentzian FWHM, placing this system in the true continuum-coupling limit; quantum survival tracks the classical envelope exactly. *Centre:* Cyclopropane ($\Gamma_{\text{IVR}} = 8.0 \text{ cm}^{-1}$, $\rho = 1.27 \text{ st/cm}^{-1}$, $\chi_0 = 0.0037$, Tier 2): Γ_{IVR} is RRKM-inferred; the sparse state density produces visible quantum recurrences at short times, but the envelope follows $\exp(-\Gamma t)$ and the RRKM thermal ensemble recovers the smooth rate. *Right:* CH_3NC ($\Gamma_{\text{IVR}} = 1.77 \text{ cm}^{-1}$, $\rho = 0.28 \text{ st/cm}^{-1}$, $\chi_0 = 0.0019$, Tier 2): the sparsest state density of the three; quantum recurrences are most prominent, but the RRKM thermal ensemble average recovers the $\exp(-\Gamma t)$ envelope. The χ_0 values in all three panels are identical to the main-text atlas Table 1 values, confirming self-consistency of the Fermi’s Golden Rule calibration.

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